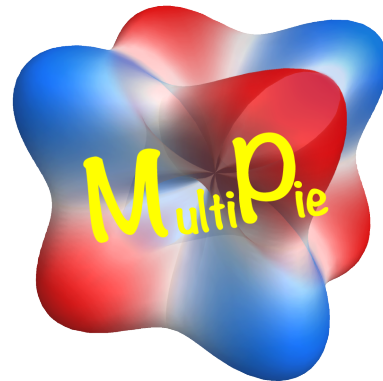


MultiPie Manual



<https://cmt-mu.github.io/MultiPie/>

for Version 2.3 (2026.7)

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Features

• Group-theoretical data

content	point group	space group	magnetic PG	magnetic SG
symmetry operation	✓	✓	✓	✓
Wyckoff (site)	✓	✓	✓	✓
Wyckoff (bond)	✓	✓		
character	✓			
harmonics	✓			
active multipole			✓	

• Symmetry Adapted Multipole Basis (SAMB) data

content	point group	space group	magnetic PG	magnetic SG
PG Clebsch-Gordan coefficient	✓			
atomic SAMB (X)	✓			
site/bond-cluster SAMB (Y)	✓	✓		
combined SAMB (Z)	✓	✓		

• List of group-theoretical data (PDF)

https://cmt-mu.github.io/MultiPie/src/group_doc.html

• Model construction and physical quantity calculations using SAMB

`$ mp_create ./input_file.py` Generate SAMB from input model file

`$ mp_analyze ./control_file.py` Compute physical quantities from control file

Physical degrees of freedom (classified by the eigen values of the following basic symmetry operations)

- Time-reversal parity (electric or magnetic)
- Spatial-inversion property (polar or axial)
- Spatial anisotropy (rank and component)

Time/Space	polar	axial
even	Q (electric)	G (electric toroidal)
odd	T (magnetic toroidal)	M (magnetic)

$$X_{lm} (X = Q, G, T, M)$$

Point groups are subgroups of the rotation group

Unitary trans.

classified by PG irreps.
(Γ, n): irrep. and multiplicity
 γ : component

$$X_{lm} \rightarrow X_{l\Gamma n\gamma} = \sum_m c_{\Gamma n\gamma}^{lm} X_{lm}$$

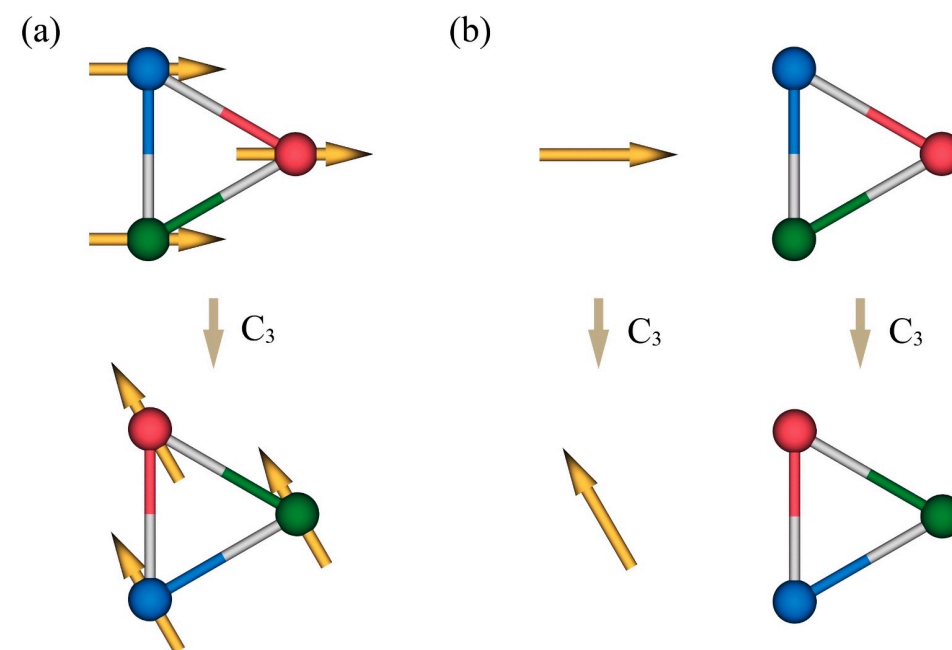
PG symmetry op. acts independently to internal d.o.f. and structure

- internal d.o.f. (atomic multipole) X_{lm}
- structure (cluster multipole) Y_{lm}
- combined basis (combined multipole)

$$Z_{lm} = \sum_{l_1 m_1, l_2 m_2} c_{lm}^{l_1 m_1, l_2 m_2} X_{l_1 m_1} Y_{l_2 m_2}$$



$$\Rightarrow \{ Z_j \}$$



symmetry operation

internal d.o.f. (atomic) structure (cluster)

MultiPie - Model Construction (Example : Graphene)

Input file

```
# graphene_in.py
graphene_in = {
    "model": "graphene", # name of model.
    "group": 191, # No. of space group.
    "cell": {"c": 4}, # set large enough interlayer distance.
    #
    "site": {"C": ("[1/3,2/3,0]", "pz")}, # positions of C site and its orbital.
    "bond": [{"C", "C", [1, 2]}], # C-C bonds up to 2nd neighbors.
    #
    "spinful": False, # spinless.
    #
    "SAMB_select": {"X": []}, Q, G, T, M (all types)
}
```

Construction \$ mp_create ./graphene_in.py

Model output ./graphene/graphene.pdf

Table 2: Harmonics **symmetry of SAMB**

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
1	$Q_0(A_{1g})$	A_{1g}	0	Q, T	-	-	1
2	$G_1(A_{2g})$	A_{2g}	1	G, M	-	-	z
3	$Q_3(B_{1u})$	B_{1u}	3	Q, T	-	-	$\frac{\sqrt{10}y(3x^2-y^2)}{4}$
4	$Q_3(B_{2u})$	B_{2u}	3	Q, T	-	-	$\frac{\sqrt{10}x(x^2-3y^2)}{4}$
5	$Q_{1,1}(E_{1u})$	E_{1u}	1	Q, T	-	1	x
6	$Q_{1,2}(E_{1u})$					2	y

continued ...

rank, component
(irrep.)

anisotropy

Structure of model `./graphene/graphene.qtdw`

C site (representative)
 p_z orbital only

C-C hopping
up to 1st and second neighbors

arrow represents
definition of positive
bond direction

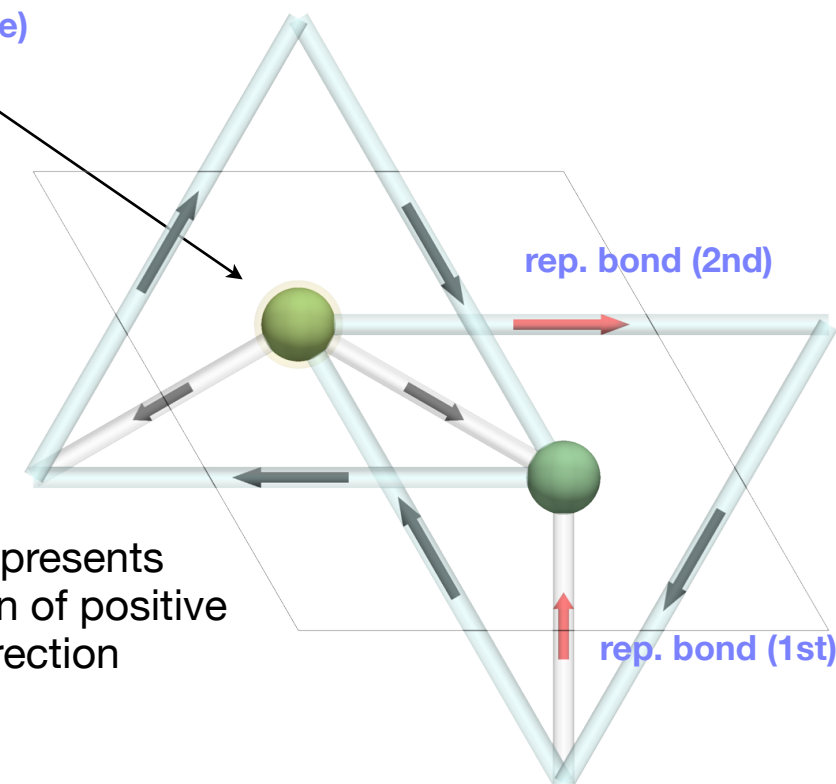


Table 3: dimension = 2 **basis of Hamiltonian matrix**

#	orbital@atom(SL)	#	orbital@atom(SL)
0	$ p_z\rangle @C(1)$	1	$ p_z\rangle @C(2)$

Table 4: Atomic basis (orbital part only)

orbital	definition
$ p_z\rangle$	z

definition of atomic basis

In the spinful case
 $x \rightarrow (x, \uparrow), (x, \downarrow)$ etc.

MultiPie - Model (Example : Graphene)

Basis of site cluster

Table 7: 'C' (#1) site cluster (2c), $-6m2$ Wyckoff (site) and site symmetry

SL	position ($\mathbf{s}$)	mapping
1	[0.33333, 0.66667, 0.00000]	[1,2,3,10,11,12,16,17,18,19,20,21]
2	[0.66667, 0.33333, 0.00000]	[4,5,6,7,8,9,13,14,15,22,23,24]

fractional coordinate and mapping of symmetry op.

Basis of bond cluster (1st neighbor)

Table 8: 1-th 'C'-'C' [1] (#1) bond cluster (3a@3f), ND, $|\mathbf{v}| = 0.57735$ (cartesian) Wyckoff (bond@site) and directionality, distance of bond

SL	vector ($\mathbf{v}$)	center ($\mathbf{c}$)	mapping	head	tail	$\mathbf{R}$ (primitive)
1	[0.33333, 0.66667, 0.00000]	[0.50000, 0.00000, 0.00000]	[1,-4,-8,11,-13,16,20,-23]	(2,1)	(1,1)	[0,-1,0]
2	[-0.66667,-0.33333, 0.00000]	[0.00000, 0.50000, 0.00000]	[2,-5,-7,10,-14,17,19,-22]	(2,1)	(1,1)	[1,0,0]
3	[0.33333,-0.33333, 0.00000]	[0.50000, 0.50000, 0.00000]	[3,-6,-9,12,-15,18,21,-24]	(2,1)	(1,1)	[0,0,0]

fractional coordinate and mapping

(sublattice, +trans.)

$\mathbf{R} = \mathbf{R}_{\text{ket (tail)}} - \mathbf{R}_{\text{bra (head)}}$

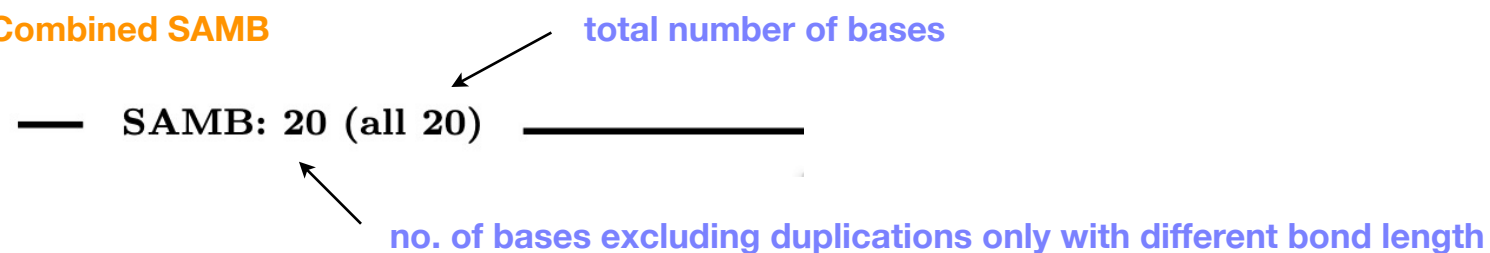
Basis of bond cluster (2nd neighbor)

Table 9: 2-th 'C'-'C' [1] (#2) bond cluster (6b@6l), ND, $|\mathbf{v}| = 1.0$ (cartesian)

SL	vector ($\mathbf{v}$)	center ($\mathbf{c}$)	mapping	head	tail	$\mathbf{R}$ (primitive)
1	[1.00000, 0.00000, 0.00000]	[0.83333, 0.66667, 0.00000]	[1,-11,16,-20]	(1,1)	(1,1)	[-1,0,0]
2	[0.00000, 1.00000, 0.00000]	[0.33333, 0.16667, 0.00000]	[2,-10,17,-19]	(1,1)	(1,1)	[0,-1,0]
3	[-1.00000,-1.00000, 0.00000]	[0.83333, 0.16667, 0.00000]	[3,-12,18,-21]	(1,1)	(1,1)	[1,1,0]
4	[-1.00000, 0.00000, 0.00000]	[0.16667, 0.33333, 0.00000]	[4,-8,13,-23]	(2,1)	(2,1)	[1,0,0]
5	[0.00000,-1.00000, 0.00000]	[0.66667, 0.83333, 0.00000]	[5,-7,14,-22]	(2,1)	(2,1)	[0,1,0]
6	[1.00000, 1.00000, 0.00000]	[0.16667, 0.83333, 0.00000]	[6,-9,15,-24]	(2,1)	(2,1)	[-1,-1,0]

MultiPie - Model (Example : Graphene)

Combined SAMB



• C : 'C' site-cluster name of site cluster

- * bra: $\langle p_z |$ basis of atomic SAMB
- * ket: $|p_z\rangle$
- * wyckoff: 2c Wyckoff (site)

$$\boxed{z1} \quad Q_0^{(c)}(A_{1g}) = Q_0^{(a)}(A_{1g})Q_0^{(s)}(A_{1g})$$

expression of combined SAMB
atomic x site-cluster

$$\boxed{z5} \quad Q_3^{(c)}(B_{1u}) = Q_0^{(a)}(A_{1g})Q_3^{(s)}(B_{1u})$$

ID of basis

• C;C_001_1 : 'C'-'C' bond-cluster name of bond cluster (tail; head_neighbor_multiplicity)

- * bra: $\langle p_z |$ basis of atomic SAMB
- * ket: $|p_z\rangle$
- * wyckoff: 3a@3f Wyckoff (bond@site)

$$\boxed{z2} \quad Q_0^{(c)}(A_{1g}) = Q_0^{(a)}(A_{1g})Q_0^{(b)}(A_{1g})$$

expression of combined SAMB
atomic x bond-cluster

$$\boxed{z6} \quad T_3^{(c)}(B_{1u}) = Q_0^{(a)}(A_{1g})T_3^{(b)}(B_{1u})$$

$$\boxed{z9} \quad T_{1,1}^{(c)}(E_{1u}) = \frac{\sqrt{2}Q_0^{(a)}(A_{1g})T_{1,1}^{(b)}(E_{1u})}{2}$$

$$\boxed{z10} \quad T_{1,2}^{(c)}(E_{1u}) = \frac{\sqrt{2}Q_0^{(a)}(A_{1g})T_{1,2}^{(b)}(E_{1u})}{2}$$

matrix of atomic SAMB

- bra: $\langle p_z |$ basis of atomic SAMB
- ket: $|p_z\rangle$

$$\boxed{x1} \quad Q_0^{(a)}(A_{1g}) = \begin{bmatrix} 1 \end{bmatrix} \quad 1 \times 1 \text{ 行列}$$

vector of site and bond cluster

** Wyckoff: 2c Wyckoff (site)

$$\boxed{y1} \quad Q_0^{(s)}(A_{1g}) = \begin{bmatrix} \frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2} \end{bmatrix} \quad 2\text{-site cluster}$$

$$\boxed{y2} \quad Q_3^{(s)}(B_{1u}) = \begin{bmatrix} \frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2} \end{bmatrix}$$

** Wyckoff: 3a@3f Wyckoff (bond@site)

$$\boxed{y3} \quad Q_0^{(b)}(A_{1g}) = \begin{bmatrix} \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3} \end{bmatrix} \quad 3\text{-bond cluster}$$

$$\boxed{y4} \quad T_3^{(b)}(B_{1u}) = \begin{bmatrix} \frac{\sqrt{3}i}{3}, \frac{\sqrt{3}i}{3}, \frac{\sqrt{3}i}{3} \end{bmatrix}$$

$$\boxed{y5} \quad T_{1,1}^{(b)}(E_{1u}) = \begin{bmatrix} 0, -\frac{\sqrt{2}i}{2}, \frac{\sqrt{2}i}{2} \end{bmatrix}$$

...

Control file

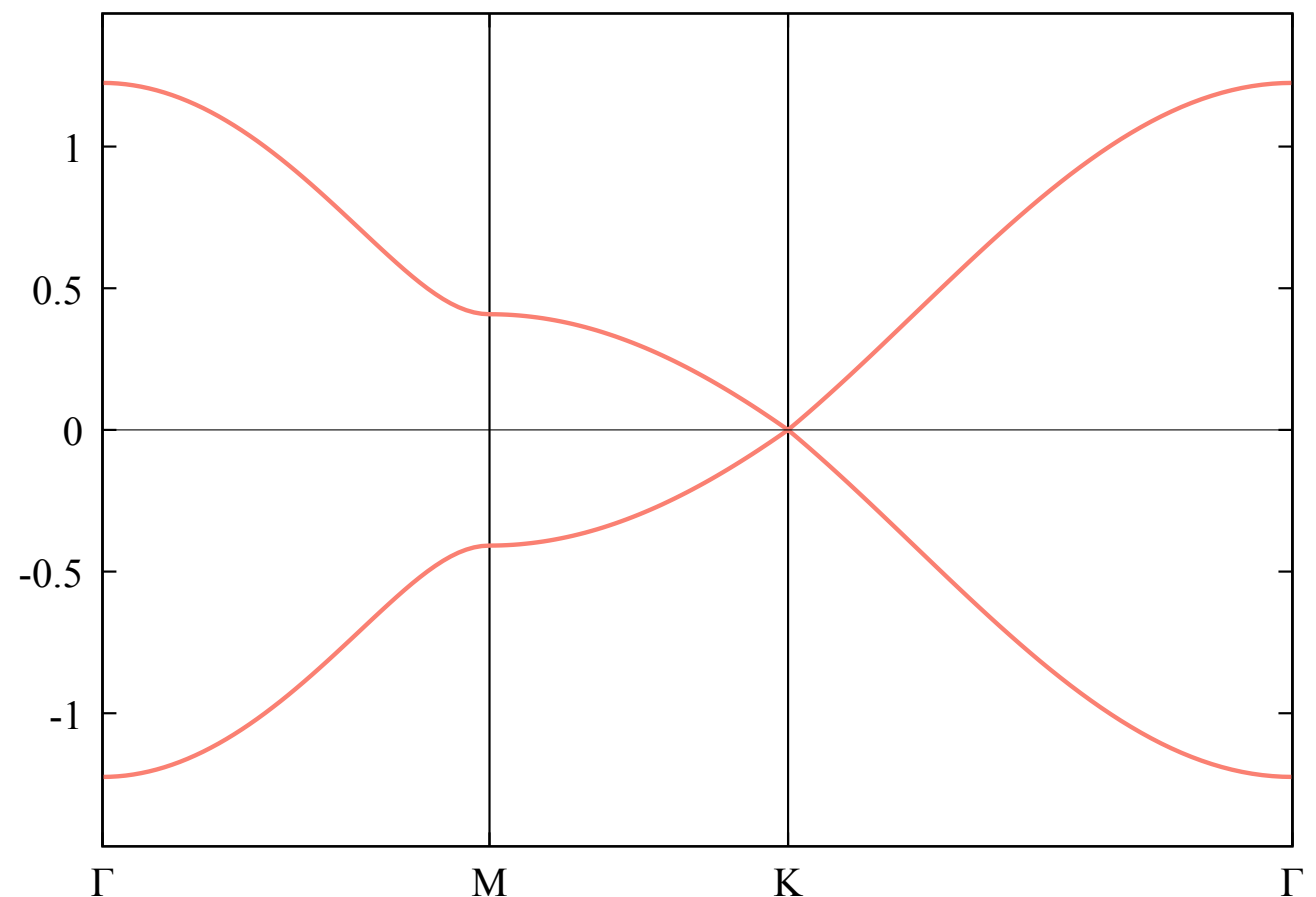
```
# graphene_ctrl.py
graphene_ctrl = {
    "samb": {
        "model": "graphene",    model name
        "parameter": {
            "z2": 1.0,          ID and weight of SAMBs used
        },
        "samb_figure": True,    generate QtDraw file for SAMB ?
    },
    "output": {
        "dispersion": {
            "k_point": {"Γ": "[0,0,0]", "K": "[1/3,1/3,0]", "M": "[1/2,0,0]"},    definition of k point
            "k_path": "Γ-M-K-Γ",          high-symmetry line ( connected with "-", discontinuities with "|" )
        },
    },
}
```

Analyze & Output

(Now developing step by step)

```
$ mp_analyze ./graphene_ctrl.py
```

band dispersion [./graphene/output/graphene_dispersion.pdf](#)



MultiPie - Matrix Element (Example : Graphene)

Matrix-element file `./graphene/graphene_matrix.py`

```
graphene = {
    "model": "graphene",
    "source": "graphene.pkl",
    "date": "2026-07-12 10:23:48",
    "select": {      filter conditions for selecting basis
        "X": ["Q", "G", "M", "T"],
        "l": [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11],
        ...
    },
    "dimension": 2,  dimension of matrix
    "ket_site": {
        "pz@C(1)": [0.3333333333333333, 0.6666666666666666, 0.0],
        "pz@C(2)": [0.6666666666666666, 0.3333333333333333, 0.0]
    },
    "index": {("C", 1, 1): (0, 1), ("C", 2, 1): (1, 1)},  (atom, sublattice, orbital rank) → (top of block matrix, size of block matrix)
    "vector": {      correspondence between cluster and bond vector (primitive cell)
        "C": [[0.0, 0.0, 0.0]],
        "C;C_001_1": [[0.33333333, 0.66666666, 0.0], [-0.66666666, -0.33333333, 0.0], [0.33333333, -0.33333333, 0.0]],
        ...
    },
    "cluster": {      correspondence between ID and cluster
        "z1": "C",
        "z2": "C;C_001_1",
        ...
    },
    "matrix": {      ID : { (n1, n2, n3, m, n) : (matrix element, bond_no), ... }
        "z1": {(0, 0, 0, 0, 0): ("sqrt(2)/2", 0), (0, 0, 0, 1, 1): ("sqrt(2)/2", 0)},
        "z2": {(0, -1, 0, 1, 0): ("sqrt(6)/6", 1), (1, 0, 0, 1, 0): ("sqrt(6)/6", 2), ...},
        ...
    }
}
```

matrix element in momentum space

$$\mathbf{k} = 2\pi(\kappa_1 \mathbf{b}_1 + \kappa_2 \mathbf{b}_2 + \kappa_3 \mathbf{b}_3) \quad \text{fractional k vector (primitive cell)}$$

$$[\mathbb{Z}_j(\mathbf{k})]_{m,n} = \sum c(n_1, n_2, n_3, m, n) e^{2\pi i \kappa \cdot (\mathbf{R} - \mathbf{r}_m + \mathbf{r}_n)}$$

$$\mathbf{R} = \mathbf{R}_n - \mathbf{R}_m = n_1 \mathbf{a}_1 + n_2 \mathbf{a}_2 + n_3 \mathbf{a}_3$$

(primitive cell)

frac. coord.
in primitive unit cell

e.g.

```
position = list( graphene["ket_site"].values() )
r_m, r_n = position[m], position[n]
```

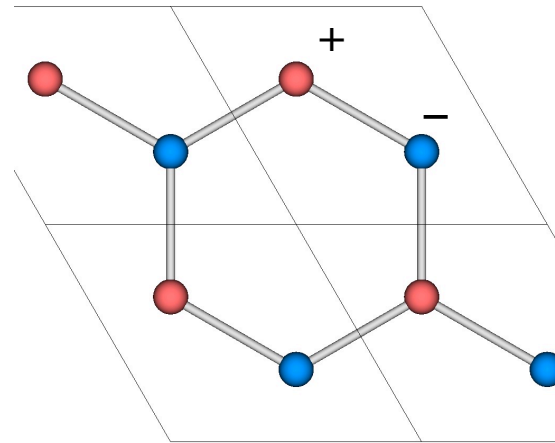

MultiPie - How to use basis (Example : Graphene)

Basis of non identity irreps.

mass term `samb/graphene_C.qtdw (y2)`

$$\boxed{\text{z5}} \quad Q_3^{(c)}(B_{1u}) = Q_0^{(a)}(A_{1g})Q_3^{(s)}(B_{1u})$$

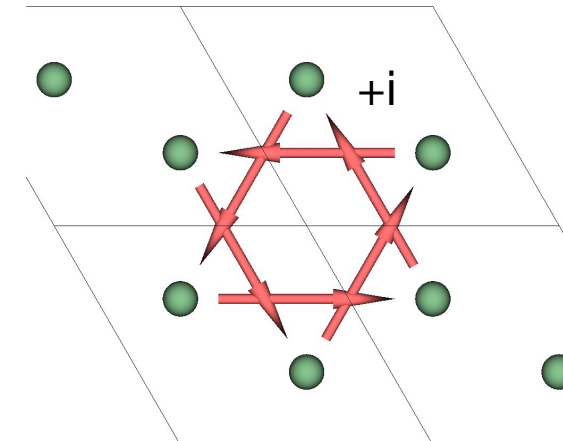
$$\boxed{\text{y2}} \quad Q_3^{(s)}(B_{1u}) = \begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{bmatrix}$$



Haldane's magnetic flux `samb/graphene_C;C_002_1.qtdw (y10)`

$$\boxed{\text{z4}} \quad M_1^{(c)}(A_{2g}) = Q_0^{(a)}(A_{1g})M_1^{(b)}(A_{2g})$$

$$\boxed{\text{y10}} \quad M_1^{(b)}(A_{2g}) = \begin{bmatrix} \frac{\sqrt{6}i}{6} & \frac{\sqrt{6}i}{6} & \frac{\sqrt{6}i}{6} & \frac{\sqrt{6}i}{6} & \frac{\sqrt{6}i}{6} & \frac{\sqrt{6}i}{6} \end{bmatrix}$$



surface Rashba SOC if model is constructed with `spinful = True`, total number of bases becomes 80 (all 80)

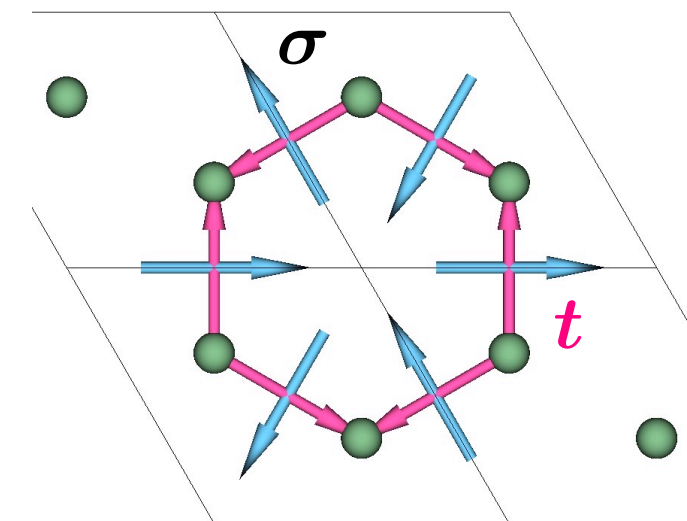
$$\boxed{\text{z12}} \quad Q_1^{(1,-1;c)}(A_{2u}) = \frac{\sqrt{2}M_{1,1}^{(1,-1;a)}(E_{1g})T_{1,2}^{(b)}(E_{1u})}{2} - \frac{\sqrt{2}M_{1,2}^{(1,-1;a)}(E_{1g})T_{1,1}^{(b)}(E_{1u})}{2}$$

$$\boxed{\text{x3}} \quad M_{1,1}^{(1,-1;a)}(E_{1g}) = \begin{bmatrix} 0 & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & 0 \end{bmatrix}$$

$$\boxed{\text{x4}} \quad M_{1,2}^{(1,-1;a)}(E_{1g}) = \begin{bmatrix} 0 & -\frac{\sqrt{2}i}{2} \\ \frac{\sqrt{2}i}{2} & 0 \end{bmatrix}$$

$$\boxed{\text{y5}} \quad T_{1,1}^{(b)}(E_{1u}) = \begin{bmatrix} 0 & -\frac{\sqrt{2}i}{2} & \frac{\sqrt{2}i}{2} \end{bmatrix}$$

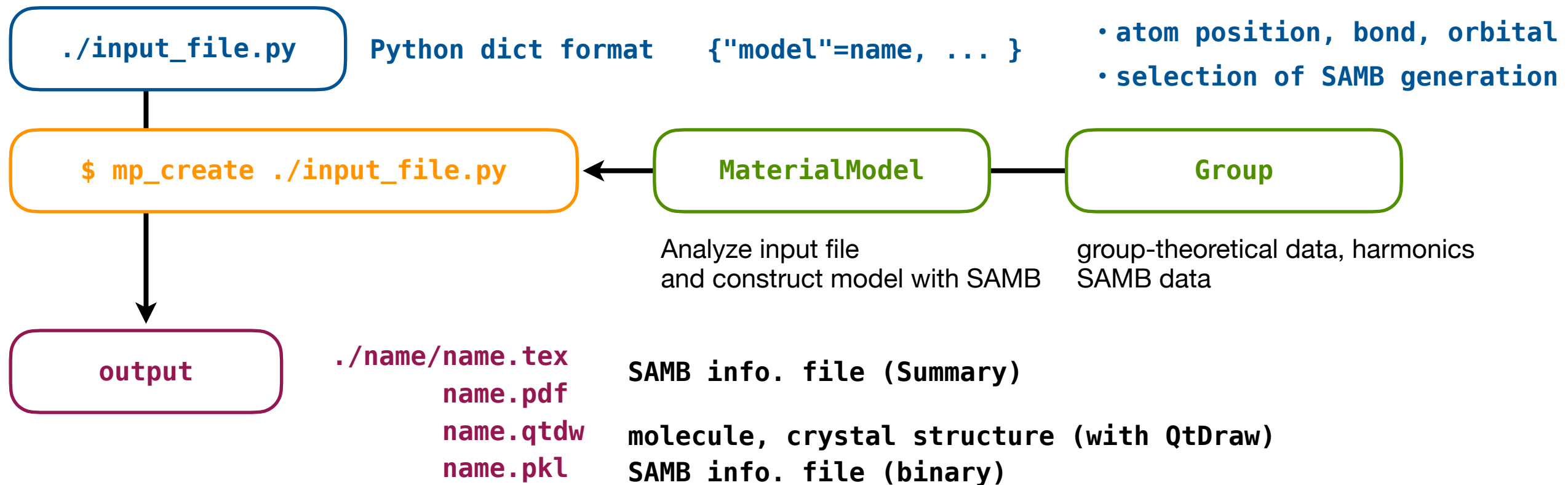
$$\boxed{\text{y6}} \quad T_{1,2}^{(b)}(E_{1u}) = \begin{bmatrix} \frac{\sqrt{6}i}{3} & -\frac{\sqrt{6}i}{6} & -\frac{\sqrt{6}i}{6} \end{bmatrix}$$



Model Generation and Physical-Quantity Calculations

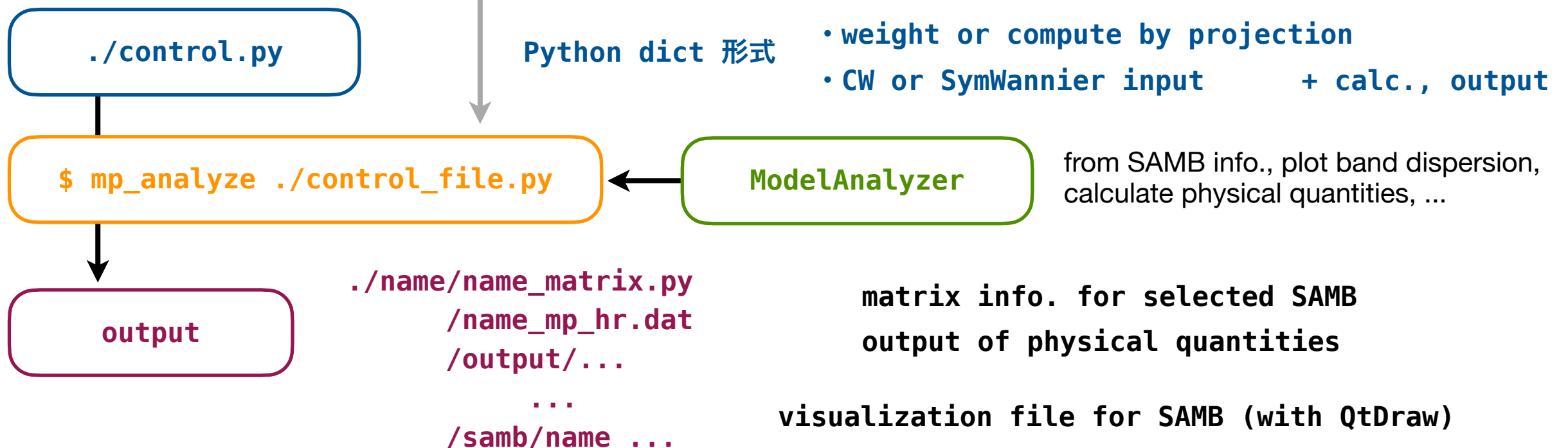
1. Model construction

```
from multipie import Group, MaterialModel
```



2. Plot band, physical quantities (Now developing step by step)

```
from multipie import ModelAnalyzer
```



MultiPie - Input File (mp_create)

Input file

default values when omitted

```
default_model = {
  "model": "unknown",    model name : used as prefix for output folders/file names
  "group": "C1",         Crystallographic PG or SG name (Shoenflies symbol or SG no.)
  "cell": {},            unit-cell parameter (a, b, c, alpha, beta, gamma)
  "spinfo": False,       spinful orbital ?
  "site": {},            rep. site in unit cell  "name": ( [frac. coord.], [orbital list] )
  "bond": [],            rep. bond in unit cell  ( "tail", "head", [neighbor list] )
  "SAMB_select": { # select combined SAMB.          filter conditions in generating SAMB
    "X": ["Q", "G"], # type, "Q/G/T/M", []=all.
    "l": [], # rank, []=all.
    "Gamma": [], # "IR"=identity, []=all, "..."/["...", "..."]=specified irreps.
    "s": [], # spin, 0/1, []=all.
  },
  "atomic_select": { # select atomic SAMB.          filter conditions in generating atomic SAMB
    "X": [], # type, "Q/G/T/M", []=all.
    "l": [], # rank, []=all.
    "Gamma": [], # "IR"=identity, []=all, "..."/["...", "..."]=specified irreps.
    "s": [], # spin, 0/1, []=all.
  },
  "site_select": { # select site-cluster SAMB.      filter conditions in generating site-cluster SAMB
    "l": [], # rank, []=all.
    "Gamma": [], # "IR"=identity, []=all, "..."/["...", "..."]=specified irreps.
  },
  "bond_select": { # select bond-cluster SAMB.      filter conditions in generating bond-cluster SAMB
    "X": [], # type, "Q/T/M", []=all.
    "l": [], # rank, []=all.
    "Gamma": [], # "IR"=identity, []=all, "..."/["...", "..."]=specified irreps.
  },
  ...
}
```

MultiPie - Input File (mp_create)

```

...

"toroidal_priority": False,    in generating SAMB, priority for toroidal multipoes ?
"max_neighbor": 10,    # for DFT fitting, e.g. =200.    max. of neighbor in searching bond
"search_cell_range": (-2, 3, -2, 3, -2, 3),    max. range of cell in searching bond
    # (a1, a2, a3) range. for DFT fitting, e.g.=(-10, 10, -10, 10, -10, 10).
"qtdraw": {    option in generating QtDraw files
    "create": True,    # create QtDraw file ?    generate ?
    "mode": "standard",    # mode, "standard/detail".    standard or detail ?
    "view": None,    # [a,b,c], if None, default=[6,5,1].    default view
    #
    "rep_site": True,    # show representative site ?    highlight for rep. site ?
    "rep_bond": True,    # show representative bond ?    highlight for rep. bond ?
    #
    "max_neighbor": 5,    # max. neighbor to draw.    max. neighbor to plot bonds
    "cell_mode": None,    # "off/single/all", if None, single for SG, off for PG.    cell display mode
    "scale": None,    # default or magnification scale.    scale of site/bond size and width ?
    "site_radius": 0.07,    # base site radius.    default radius of site
    "bond_width": 0.02,    # base bond width.    default widht of bond
},
"pdf": {    option in generating PDF
    "create": True,    # create PDF file ?    generaete ?
    "common_samb": True,    # display common SAMB ?    display common bond SAMB ?
    "harmonics": True,    # display harmonics ?    display harmonics info. ?
    "max_neighbor": 5,    # max. neighbor to write (None: all).    max. of neighbor
},
}

```

MultiPie - Control File (mp_analyze)

Control file default values when omitted

```
default_control = {
  "mode": "samb",   mode: samb (SAMB only), wannier (Wannier only), symcw (SAMB and wannier)
  "samb": {
    "model": "unknown",  model name or None
    "select": {          select condition of SAMB          S=site name, R=orbital rank, N=neighbor bond
      # "site": [("A", [1,2])],  list of S or (S,[R]) format
      # "bond": [[0,1,2]],      list of S1;S2, [N], (S1;S2,[N]), (S1;S2,R1;R2,[N]) format
      # "X": [],               list of SAMB type, Q/G/T/M ([]=all types)
      # "l": [],               list of SAMB rank, 0, 1, ..., 11 ([]=all ranks)
      # "Gamma": ["A1g"],      list of SAMB irrep. ("IR"=identity irrep., []=all irreps.)
      # "s": [],               list of SAMB internal rank, 0,1 ([]=0 and 1)
    },
    "parameter": {       finite weight (float or sympy) in case of empty dict. and given wannier, compute by projection
      # "z1": 1.0,        In case of file name, read "./name/filename"
    },
    "samb_figure": False, output QtDraw files of SAMB ?
    "k_multipole": False, create momentum rep. ?
    "NG_sum_rule": False, enforce NG sum rule ?
  },
  "wannier": {           Wannier info. from DFT → Wannier computation
    "seedname": None,    read seedname +(.win, .nnkp, .hr.dat/hr_cw.dat)
    "ket_wannier": [],   Wannier ket list [site-cluster, sublattice, orbital], auto detection for empty
  },
  "output": {            info. for computing physical quantities
    "dir": "output",     output folder
    "fourier": { "tb_gauge": True }, use atomic position for phase (TB gauge) ?
    "dos": False,        compute DOS ?
    "dispersion": {      for band dispersion
      "k_path": "",       high-sym. line, with "-", discont. with "|" [for "", use default path, k_point is generated]
      "k_point": {"Γ": "[0,0,0]"}, definition of k-point (primitive fractional)
      "local": [],        expectation values of local operators (Sx,Sy,Sz, Lx,Ly,Lz, Qu,Qv,Qyz,Qxz,Qxy)
      "z": [],            SAMB ID to compute expectation value (Not implemented)
    },
  },
}
```


MultiPie - コントロールファイル (mp_analyze)

Control file

default values when omitted

```
default_control = {
    "samb": {
        "model": "unknown",  model name or None
        "select": {  select condition of SAMB: S=site name, R=orbital rank, N=neighbor bond
            # "site": [("A", [1,2])],  list of S or (S,[R]) format
            # "bond": [[0,1,2]],  list of S1;S2, [N], (S1;S2,[N]), (S1;S2,R1;R2,[N]) format
            # "X": [],  list of SAMB type: Q,G,T,M ([]=all types)
            # "l": [],  list of SAMB rank: 0,1,...,11 ([]=all ranks)
            # "Gamma": ["A1g"],  list of SAMB irrep.: ("IR"=identity irrep., []=all irreps.)
            # "s": [],  list of SAMB internal rank: 0,1 ([]=0 and 1)
        },
        "parameter": {  finite weight (float/sympy) empty+wannier, compute by projection
            # "z1": 1.0,  In case of file name, read "./name/filename"
        },
        "samb_figure": False,  output QtDraw files of SAMB ?
        "k_multipole": False,  create momentum rep. ?
        "NG_sum_rule": False,  adopt Nambu-Goldstone sum rule ?
    },
    "wannier": {  Use DFT→Wannier based info.
        "cw": None,  info file name not implemented yet
    },
    "output": {  info. for computing physical quantities
        "dir": "output",  output folder
        "fourier": { "tb_gauge": True },  use atomic position for phase (TB gauge) ?
        "dos": False,  compute DOS ?
        "dispersion": {  for band dispersion
            "power": None,  power for plot [None=1]
            "k_path": "",  high-symmetry path: connect by "-", discontinuity "|" [empty=default k_point generated)
            "k_point": {"Γ": "[0,0,0]"},  definition of k point (primitive,fractional)
            "local": [],  expectation values of local operators (Sx,Sy,Sz,Lx,Ly,Lz,Qu,Qv,Qyz,Qxz,Qxy)
            "z": [],  SAMB ID to compute expectation value not implemented yet
        },
    },
}
```

Group Class

Initialization Generate crystallographic point group, space group, magnetic PG, and magnetic SG

Group(tag, with_opt=False)

tag

example

• point group	"PG:no", "Schoenflies"	no = 1 - 47	"PG:5", "C3v"
• space group	"SG:no", "Schoenflies", no	no = 1 - 230	"SG:152", "D3^2", 152
• magnetic PG	"MPG:HM_id"	HM_id = PG.no.ID	"MPG:12.3.8"
• magnetic SG	"MSG:BNS_id"	BNS_id = SG.no	"MSG:152.3"

with_opt

load additional data ?

group.opt[key]

dict key	content
harmonics_multipole	multipolar harmonics (dipole, quadrupole, octupole)
harmonics_irrep	harmonics belonging to irrep. (irrep. → rank, no)
representation_matrix	representation matrix

Global info. **Group.global_info()**

dict key	content	dict key	content
id_set	list of tag classified by category	harmonics	max. of rank, name, definition, etc.
tag	unique tag → conventional tag	response_tensor	definition of tensor and multipole
id	conventional tag → unique tag	root_cluster	misc. of cluster SAMB
character	alias and compatibility relation		

Detail of API

https://cmt-mu.github.io/MultiPie/src/api_core/group.html

MaterialModel Class

MaterialModel - Overview

Initialization Create model from structure, symmetry, and atomic orbital info.

MaterialModel(**topdir**=None, **verbose**=False)

topdir top working directory (current dir when omitted)

verbose display log info. ?

Dict data https://cmt-mu.github.io/MultiPie/src/api_core/material_model.html

Property **MaterialModel.group**

Function

function name	content	function name	content
x	get atomic SAMB	y	get cluster SAMB
z	get combined SAMB	analyze	analyze input data
cell_site_primitive	get site position in primitive cell	clear	clear info.
get_atomic_samb	get all of atomic SAMB	get_cluster_samb	get all of cluster SAMB
get_combined_samb	get all of combined SAMB	get_combined_samb_matrix	get combined SAMB matrix
get_hr	get Hamiltonian matrix	get_ket_site	get ket and corresponding site
get_multipole_expression	get expression of multipole	get_samb_matrix	get full matrix info.
ket	list of ket basis in LaTeX format	load	load model file (binary)
save	save model info. in binary	save_pdf	save model info. in PDF
save_samb_hr	save SAMB matrix in hr format	save_samb_matrix	save SAMB matrix
save_samb_qtdraw	save QtDraw for SAMB	save_view	save structure file

Detail of API

https://cmt-mu.github.io/MultiPie/src/api_core/material_model.html

ModelAnalyzer Class

Initialization Compute and output physical quantities from model info.

ModelAnalyzer(**N1=50**, **N2=50**, **N3=50**, **topdir=None**, **verbose=False**)

N1, N2, N3 no. of divisions in 1st Brillouin zone (primitive cell)

topdir top working directory (current dir when omitted)

verbose display log info. ?

Property

property name	content	property name	content
HR	Hamiltonian matrix (real space)	model	MaterialModel object
name	model name	output	control (output related)
samb	control (samb related)	wannier	control (wannier related)

Function

analyze(**control**) read control file, compute physical quantities, and output

control control file name or dict.

set_grid_size(**N1**, **N2**, **N3**) set no. of divisions

N1, N2, N3 no. of divisions in 1st Brillouin zone (primitive cell)

Detail of API

https://cmt-mu.github.io/MultiPie/src/api_core/model_analyzer.html